Some Fuzzy Measures Constructing Paradigm and Methods From Given Measure Spaces and Evaluation Information

LeSheng Jin, *Senior Member, IEEE*, Radko Mesiar, and Ronald R. Yager, *Life Fellow, IEEE*

***Abstract*—With some theoretical derivation and practical decision-making problems, in this article, we propose a constructing paradigm for generating fuzzy measures in some certain decision situations, such as in risk evaluation and decision making. The constructing paradigm involves four constructing factors: source measure space, construction function, construction choice, and object/constructed measure space. We use some strict and consistent formulation to reinvent the processes of constructing the well-known belief measure and plausibility measure. In this a, the belief measure and plausibility measure are regarded as the two extremities of some pessimism–optimism preferences, and we further propose two construction choices to construct the families of fuzzy measures to generalize and seamlessly connect them. We also propose three different ways to determine construction functions in some certain decisional situations. Several relevant mathematical properties and numerical examples are also presented.**

***Index Terms*—Aggregation operators, decision making, evaluation, fuzzy measure constructing, fuzzy measures.**

I. Introduction

Over the past few decades, the theories and techniques of information fusion have well developed in different areas, including the evaluation for systems reliability [1], image processing [2], [3], scientomatics [4], [5], approximate evaluations [6], decision analysis [7], and decision making [8]–[11], as well as some wider applications in different main subjects, such as economics [12], [13] and geography science [14]. Next, we review some popular and powerful aggregation techniques revolving around aggregation operators and fuzzy integrals.

A. Aggregation Operators, Fuzzy Measure, and Fuzzy Integrals

Aggregation operators (also termed as aggregation functions) serve as a very important theoretical underpinning for information fusion. Some commonly known aggregation operators include mean, median, maximum, minimum, *t*-norms/*t*-conorms, weighted averaging operators, and ordered weighted averaging (OWA) operators [11].

Real practices and applications have contributed enormously to the development of aggregation operators. For example, different weights and fuzzy measures determination methods and techniques proposed majorly for multicriteria decision making (MCDM) actually diversified the research areas and scopes of aggregation operators, further promoting the combination of aggregation theory with real practice. While the analytic hierarchy process (AHP) [15] and bipolar preference weighting mechanism based on OWA operators [11] are two very typical and well-known weights and fuzzy measures determination methods; numerous other different ways and techniques [16]–[23] are also worthy of great attention.

Without the loss of generality, for any input vector $\mathbf{x} = (x_i)_{i=1}^n \in [0,1]^n$, an aggregation operator is a function $F : [0,1]^n \to [0,1]$ satisfying boundary conditions $F(\mathbf{0}) = 0$ and $F(\mathbf{1}) = 1$, and monotonicity $F(\mathbf{x}) \le F(\mathbf{y})$ whenever $x_i \le y_i$ $(i \in N = \{1,\ldots,n\})$ [24].

For $N = \{1,\ldots,n\}$, a (normalized) fuzzy measure [25] (also known as capacity [26]) on N is a special set function $m : 2^N \to [0,1]$ such that $m(\varnothing) = 0$ and $m(N) = 1$, and $m(A) \le m(B)$ whenever $\varnothing \subseteq A \subset B \subseteq N$. With a fuzzy measure, scholars have defined a number of different aggregation operators, called fuzzy integrals in general, including some very classical or well-known integrals, e.g., Choquet integral [26], Sugeno integral [25], Shilkret integral [27], Pan integral [28], and some relatively newly introduced functions involving fairly complex structures, e.g., decomposition integral [12], [29], derived Choquet integral [30], and some universal and extended integrals [31]–[33].

As the two most commonly known fuzzy integrals, in what follows, we briefly review the (discrete) Choquet integral and Sugeno integral.

For any input vector $\mathbf{x} \in [0,1]^n$, we assume the following.

1) The Sugeno integral with given fuzzy measure $m : 2^N \to [0,1]$, $\mathrm{Su}_m : [0,1]^n \to [0,1]$, is defined such that

$$\mathrm{Su}_m(\mathbf{x}) = \max_{i\in N} \left(\min\left(m(\{k \in N : x_k \ge x_i\}), x_i\right)\right). \quad (1)$$

2) The Choquet Integral with given fuzzy measure $m : 2^N \to [0,1]$, $\mathrm{Ch}_m : [0,1]^n \to [0,1]$, is defined such that

$$\mathrm{Ch}_m(\mathbf{x}) = \sum_{i=1}^{n} \left(x_{\sigma(i)} - x_{\sigma(i+1)}\right) \cdot m(\{\sigma(1),\ldots,\sigma(i)\}) \quad (2)$$

with convention $x_{\sigma(n+1)} = 0$, where $\sigma : N \to N$ can be any proper permutation such that for any $i < j$, we have $x_{\sigma(i)} \ge x_{\sigma(j)}$.

Recall that when m is a probability measure, then the Choquet integral with m in (2) is degenerated to the weighted averaging operator with (normalized) weight vector $\mathbf{w} = (m(\{i\}))_{i=1}^n$, that is, $\mathrm{Ch}_m(\mathbf{x}) = \sum_{i=1}^n m(\{i\}) \cdot x_i$.

In decision making, $N = \{1,\ldots,n\}$ often represents a collection of n criteria concerned in some certain evaluation problems, and for any subcollection of criteria $A \subseteq N$, $m(A)$ indicates the importance of all the criteria in that subcollection as a combined effect. Meanwhile, any input vector $\mathbf{x} = (x_i)_{i=1}^n \in [0,1]^n$ represents the collection of values, scores, payoffs, the extents of positive or negative effect, etc., corresponding to those n criteria, and the aggregation result, such as

Manuscript received April 14, 2020; revised November 8, 2020; accepted December 4, 2020. Date of publication December 14, 2020; date of current version March 1, 2022. This work was supported in part by the Scientific Research Startup Foundation under Grant 184080H202B165, in part by the Science and Technology Assistance Agency under Contract APVV-17-0066, and in part by the project of Grant Agency of the Czech Republic under Grant 18-06915S. *(Corresponding author: LeSheng Jin.)*

LeSheng Jin is with the Business School, Nanjing Normal University, Nanjing 210023, China (e-mail: jls1980@163.com).

Radko Mesiar is with the Faculty of Civil Engineering, Slovak University of Technology, 811 07 Bratislava, Slovakia, and also with the Faculty of Science, Department of Algebra and Geometry, Palacky University Olomouc, 771 47 Olomouc, Czechia (e-mail: radko.mesiar@stuba.sk).

Ronald R. Yager is with the Machine Intelligence Institute, Iona College, New Rochelle, NY 10801 USA (e-mail: yager@panix.com).

Digital Object Identifier 10.1109/TFUZZ.2020.3044247

1063-6706 © 2020 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Xi'an Jiaotong-Liverpool University. Downloaded on July 23,2025 at 16:35:24 UTC from IEEE Xplore. Restrictions apply.

$\mathrm{Ch}_m(\mathbf{x})$ and $\mathrm{Su}_m(\mathbf{x})$, is considered to be a comprehensive and rigorous evaluation result.

With long-time efforts, significant signs of progress have been made in the theories and applications of aggregation operators [24], [34]–[36] and likewise in the areas of fuzzy measures and fuzzy integrals [37]–[39].

B. Importance and Some Developments of Fuzzy Measure Constructing

Fuzzy integrals can perfectly generalize some important types of aggregation operators. For example, the weighted averaging operators are special cases of Choquet integrals with probability measures, and OWA operators are the special cases of Choquet integrals with symmetrical fuzzy measures [24]. Such a generalization is only one aspect that makes the studies of fuzzy integrals become the lasting hot areas of research. Some other advantages of fuzzy integrals can be found in practices. Recall that using different types of fuzzy measures, decision makers can model some general phenomena in preferences modeling, such as the effect of criteria synergy, which otherwise is hard to be modeled merely using weight vectors.

Fuzzy integrals and related applications are of significant importance and well developed, whereas the constructing methods of desired fuzzy measures have not been as deeply studied as the fuzzy integrals themselves and their applications. In actual, researchers should take more attention and concern to the fuzzy measures constructing methods. Without first determining fuzzy measures, those above-mentioned fuzzy integrals cannot be practically applied in practice.

While the existing methods and theories have yet to become fruitful, some relevant research articles have still been taken slowly. For example, more theoretically, Ralevic and Paunovic [40] proposed the *c*-credibility measure. More practically, Li *et al*. [16] devised some methods for determining fuzzy measures from the tolerance perspective in MCDM. Recently, Jin *et al*. [19] proposed an interesting method to generate a desired fuzzy measure from a given probability measure and a given OWA weight vector (a subjective bipolar optimism/pessimism preference), allowing the decision makers to consider the two important types of preferences at the same time. In addition, Zhu *et al*. [41] found some practical ways to generate fuzzy measures with given parameters provided by decision makers.

The speed of the development of fuzzy measures constructing methods and theories is relatively slow in comparison with that of the development of the theories and applications of fuzzy integrals. Against this background, further studying on such constructing areas is always important and necessary. In this study, we will propose some strict and novel constructing methods, which have both theoretical derivation and practical meanings. Different from some other purely theoretical methods to construct fuzzy measures (such as the *c*-credibility measure), the proposed methods in this work have relations to the practical decision-making backgrounds, such as the situation where multiple experts are invited whose individual opinions will be considered and affect the construction processes and results. Another feature of the proposed methods lies in which some paradigms of the constructing methods will be formulated by which much more different constructing results and instances from different decision makers can be flexibly made. Moreover, some novel sets based structures and strict formulations of some probability information extraction or transformation methods can be found in the processes of fuzzy measure constructions.

The resy of this article is organized as follows. In Section II, we first propose a constructing paradigm based on both theoretical and practical motivations, and then by the paradigm, we reinvent the processes of constructing the well-known belief measure and plausibility measure. In Section III, we construct the two families of fuzzy measures that can seamlessly connect the belief measure and plausibility measure. Section IV discusses and analyzes three different methods to determine construction functions in some certain decisional scenarios. Finally, Section V concludes this article.

II. Some Theoretical Motivations and Practical Derivations for a Constructing Paradigm

This section first analyzes some basic methods with motivations to construct measures and then discusses some further generalization.

A. Basic Constructing Motivations From Both Probability and Practice

In what follows, we discuss and explain two motivating aspects of some basic measure constructing methods.

1) Recall from the probability theory [42] that there is a commonly known method to generate a new probability measure (space) from a given probability measure (space). Let $(\Omega, \mathcal{F}, P)$ be a probability space and $\xi : \Omega \to \mathbb{R}$ be a random variable, then ξ gives rise to a new probability measure (or probability space $(\mathbb{R}, \mathcal{B}(\mathbb{R}), P_\xi)$) $P_\xi : \mathcal{B}(\mathbb{R}) \to [0, 1]$ such that $P_\xi(A) = P(\omega \in \Omega : \xi(\omega) \in A)$, where $\mathcal{B}(\mathbb{R})$ is the $\sigma -$ field of Borel sets contained in the real line.

We observe that the random variable $\xi : \Omega \to \mathbb{R}$ results in a new measure P_ξ from the original given measure P. That is, some functions, such as the random variable ξ, may help generate new measures from the given measures. Hence, this motivates us to have a measure constructing method from "a given probability measure" to "the object fuzzy measure." For the given probability measure $(Q, 2^Q, P)$, where $Q = \{1, \ldots, q\}$ and 2^Q is the power set of Q (which consists of all the subsets of Q), a known mapping $\psi : Q \to N$ together with a particular constructing choice can generate a fuzzy measure space $(N, 2^N, P_\psi)$ such that for all $A \in 2^N$

$$P_\psi(A) = P(\{\omega \in Q : \psi(\omega) \in A\}). \tag{3}$$

In actual, the fuzzy measure space $(N, 2^N, P_\psi)$ constructed in this way is also a probability space (it is commonly known that a probability measure is additive and can be regarded as a special case of the normalized fuzzy measure that is not necessarily additive). Later, we will also use some similar but more complex formulations to construct different fuzzy measures (or fuzzy measure spaces) from the given probability measures (or probability spaces).

Hence, the proposed constructing models in this study have some sound theoretical derivations and sources in mathematics.

1) The abovementioned measures constructing methods from $(Q, 2^Q, P)$ to $(N, 2^N, P_\psi)$ also have derivation in the practical evaluation and decision-making problems. In risk evaluation and decision making, it involves *n* states of nature $N = \{1, \ldots, n\}$ over which we need to determine a probability distribution. In other words, we need to construct a probability measure space $(N, 2^N, P_\psi)$, and we assume that the *n* individual payoffs or evaluations with each state of nature are represented by a random variable $X : N \to [0, 1]$ such that $X(i)$ indicates the individual payoff or evaluation corresponding to the state of nature *i*. Then, a risk evaluation can be achieved by the expectation of *X*: $E(X) = \sum_{i=1}^{n} P_\psi(\{i\}) \cdot X(i)$, which, in a general sense of aggregation operator, can be also equivalently expressed by the Choquet integral of the input vector $\mathbf{x} = (x_i)_{i=1}^n = (X(i))_{i=1}^n$ with probability measure P_ψ: $\mathrm{Ch}_{P_\psi}(\mathbf{x})$.

Authorized licensed use limited to: Xi'an Jiaotong-Liverpool University. Downloaded on July 23,2025 at 16:35:24 UTC from IEEE Xplore. Restrictions apply.

The constructing is based on a known probability space $(Q, 2^Q, P)$ with $Q = \{1, \ldots, q\}$, which, for example, can represent a collection of q experts, with each of them being required to give his/her personal opinion about which one state of nature "*will occur*." In addition, $\mathbf{p} = (p_j)_{j=1}^q = (P(\{j\}))_{j=1}^q$ forms a normalized weight vector representing the relative importance among those q experts that can be obtained by different weights determination techniques, such as the AHP method and OWA weights generating techniques. The personal opinions of those experts can be well represented by the mapping $\psi : Q \to N$, as mentioned above, wherein $\psi(j) \in N$ is interpreted as expert j believing that the state of nature $\psi(j)$ will occur rather than any of the other states.

The abovementioned constructing process can be summarized into a constructing paradigm involving the following four *constructing factors*:

1) source measure space;
2) construction function;
3) construction choice;
4) object/constructed measure space.

To specify and formulated the above-discussed problem, we have the following instance:

1) source measure space: $(Q, 2^Q, P)$;
2) construction function: $\psi : Q \to N$;
3) construction choice: $P_\psi(A) = P(\{\omega \in Q : \psi(\omega) \in A\})$ for all $A \in 2^N$;
4) object/constructed measure space: $(N, 2^N, P_\psi)$.

B. Constructing Methods Using Set-Valued Functions to Generate Belief Measure and Plausibility Measure

In the preceding case, it is required that each expert must select one and only one state of nature that he/she believes "will occur," which means the occurrence of that state of nature is *exclusive* of the other states of nature. This selection is just represented by the mapping $\psi : Q \to N$.

In practice, however, it is usually possible that some invited experts are not able to predict exactly and precisely which one state of nature will occur but only able to narrow down the possibility of occurrences into a set with fewer states of nature. That is, the opinion of some expert j may correspond to a nonempty subset $\phi(j) \in 2^N \backslash \{\varnothing\}$ including one or more elements rather than necessarily including only one element belonging to N; this indicates that one of the states of nature within $\phi(j)$ will still surely occur but the exact occurring state is unknown as to the knowledge of expert j. Therefore, the range of the previous mapping "$\psi : Q \to N$" will be replaced so that we will use a new construction function $\phi : Q \to 2^N \backslash \{\varnothing\}$, which is set valued. Besides, the constructed measure space becomes $(N, 2^N, m_{P,\phi})$ and the source measure space remains still $(Q, 2^Q, P)$, with motivation similar to what has been discussed above.

Hence, one method to construct fuzzy measure $m_{P,\phi}$ is defined such that for all $A \in 2^N$

$$m_{P,\phi}(A) = P(\{\omega \in Q : \phi(\omega) \subseteq A\}). \tag{4}$$

Recall $m_{P,\phi}$ obtained by (4) is also known as a belief measure, as denoted by Bel, from the Dempster–Shafer belief theory [43], [44].

Alternatively, we can take another way to construct fuzzy measure P_ϕ such that for all $A \in 2^N$

$$m_{P,\phi}(A) = P(\{\omega \in Q : \phi(\omega) \cap A \neq \varnothing\}). \tag{5}$$

The fuzzy measure constructed in (5) is also known as a plausibility measure from the Dempster–Shafer belief theory [43], [44], as denoted by Pl.

Later, to differentiate between the notations of a constructed belief measure and a constructed plausibility measure, we denote by $m_{P,\phi,\mathrm{Bel}}$ the constructed belief measure in (4) and by $m_{P,\phi,\mathrm{Pl}}$ the constructed plausibility measure in (5), respectively.

The above-discussed constructing process for the belief measure or plausibility measure can be formulated and specified into the following instance according to the four constructing factors in the paradigm:

1) source measure space: $(Q, 2^Q, P)$;
2) construction function: $\phi : Q \to 2^N \backslash \{\varnothing\}$;
3) construction choice: $m_{P,\phi,\mathrm{Bel}}(A) = P(\{\omega \in Q : \phi(\omega) \subseteq A\})$ or $m_{P,\phi,\mathrm{Pl}}(A) = P(\{\omega \in Q : \phi(\omega) \cap A \neq \varnothing\})$ for all $A \in 2^N$;
4) object/constructed measure space: $(N, 2^N, m_{P,\phi,\mathrm{Bel}})$ or $(N, 2^N, m_{P,\phi,\mathrm{Pl}})$.

Besides, it is known that such constructed belief measure $m_{P,\phi,\mathrm{Bel}}$ (plausibility measure $m_{P,\phi,\mathrm{Pl}}$) is always superadditive (subadditive); that is, for any two disjoint subsets $A, B \in 2^N \backslash \{\varnothing\}$ $(A \cap B = \varnothing)$, $m_{P,\phi,\mathrm{Bel}}(A \cup B) \geq m_{P,\phi,\mathrm{Bel}}(A) + m_{P,\phi,\mathrm{Bel}}(B)$ $(m_{P,\phi,\mathrm{Pl}}(A \cup B) \leq m_{P,\phi,\mathrm{Pl}}(A) + m_{P,\phi,\mathrm{Pl}}(B))$.

Moreover, it is evident that the belief measures are smaller than the plausibility measures, given the same constructing conditions. That is, in the above-constructed fuzzy measures, $m_{P,\phi,\mathrm{Bel}} \leq m_{P,\phi,\mathrm{Pl}}$. In decision making, the belief measures can model some pessimistic or conservative decision attitudes, while the plausibility measures can embody some optimistic or sanguine decision attitudes. This is because if two fuzzy measures can be compared, then so can their aggregation results with the same input vector; that is, if $m_{P,\phi,\mathrm{Bel}} \leq m_{P,\phi,\mathrm{Pl}}$, then for any $\mathbf{x} \in [0,1]^n$, $\mathrm{Ch}_{\mathrm{m}_{\mathrm{P},\phi,\mathrm{Bel}}}(\mathbf{x}) \leq \mathrm{Ch}_{m_{P,\phi,\mathrm{Pl}}}(\mathbf{x})$. The equal sign in $m_{P,\phi,\mathrm{Bel}} \leq m_{P,\phi,\text{-}\circ}$ holds only when $\mathrm{card}(\phi(\omega)) = 1$ for all $\omega \in Q$, where $\mathrm{card}(A)$ indicates the cardinality of any finite set A.

The choice for the belief measures or plausibility measures provides some flexibility and is realistic in real evaluation practices.

III. Fuzzy Measures Gradually Connecting Belief Measure and Plausibility Measure

Section II illustrated and discussed how to use the constructing paradigm to regenerate the belief measure and plausibility measure with the same source measure space and construction function. In actual, when the decision problem is in terms of *profit* (not of *cost*) and evaluated using fuzzy integrals, then the belief/plausibility measure embodies a type of pessimistic/optimistic evaluation preferences. Given a preference range as between the belief measure and plausibility measure, they themselves serve as the lower boundary and upper boundary of this range. This section will propose a method to gradually connect them; that is, to find a family of fuzzy measures $\{m_{P,\phi,\mu}\}_{\mu \in \mathcal{M}^{(n)}}$, where $\mathcal{M}^{(n)}$ denotes the space of all fuzzy measures on $N = \{1, \ldots, n\}$.

Denote by μ_* the smallest fuzzy measure such that $\mu_*(A) = 0$ whenever $A \neq N$, and denote by μ^* the largest fuzzy measure such that $\mu(B) = 1$ whenever $B \neq \varnothing$. Clearly, any fuzzy measure $\mu : 2^N \to [0,1]$ satisfies $\mu_* \leq \mu \leq \mu^*$. Actually, a fuzzy measure μ can model the pessimism/optimism preference of a decision maker on the states of nature in the risk evaluation, as introduced in Section II. It is commonly known that such pessimism/optimism preference can be bipolar and be well modeled by OWA vectors [11]. Note that OWA vectors used in the aggregation functions mathematically can be also regarded as symmetrical fuzzy measures [24]. Therefore, a fuzzy measure μ can model a wider range of preferences that include bipolar preferences as modeled by OWA vectors, and a smaller/larger μ generally reflects a more pessimistic/optimistic preference to the whole risk evaluation.

Authorized licensed use limited to: Xi'an Jiaotong-Liverpool University. Downloaded on July 23,2025 at 16:35:24 UTC from IEEE Xplore. Restrictions apply.

As mentioned in Section II, in risk evaluation and decision making, we need to determine a fuzzy measure for further evaluation using aggregation functions. If we already have a wanted fuzzy measure μ that we believe can well represent the real situation and perform the evaluation, then there is no need for further operation and constructing process. However, if we are not sure such a fuzzy measure μ can exactly embody the decision situation, then we still need to consult with other experts, as discussed in Section II. Therefore, a constructing paradigm involving four constructing factors will still be used to construct a desired object/constructed measure space in which the known fuzzy measure μ will affect the process of construction choice and the result of the constructed measure space. The range of such influence will still be within the constructed belief measure $m_{P,\phi,\mathrm{Bel}}$ and plausibility measure $m_{P,\phi,\mathrm{Pl}}$, and in accordance with the magnitude of μ; that is, denote by $m_{P,\phi,\mu}$ the resulted fuzzy measure influenced by μ, then for any two fuzzy measures μ_1 and μ_2 with $\mu_1 < \mu_2$, we have $m_{P,\phi,\mu_1} \leq m_{P,\phi,\mu_2}$.

We next present two related methods. The first one is with a very simple but still meaningful structure.

Method 1: Simple method to construct a family of fuzzy measures with a given fuzzy measure μ.

This method has no more link to the constructing paradigm, and it works directly with the already constructed belief measure $m_{P,\phi,\mathrm{Bel}}$ and plausibility measure $m_{P,\phi,\mathrm{Pl}}$ and generates the family of fuzzy measures $\{m_{P,\phi,\mu}\}_{\mu\in\mathcal{M}^{(n)}}$ such that for any $A \in 2^N$

$$m_{P,\phi,\mu}(A) = (1-\mu(A)) \cdot m_{P,\phi,\mathrm{Bel}}(A) + \mu(A) \cdot m_{P,\phi,\mathrm{Pl}}(A). \tag{6}$$

Proposition 1: The set function $m_{P,\phi,\mu} : 2^N \to [0,1]$ as defined by (6) is a fuzzy measure.

Proposition 2: For the fuzzy measures $m_{P,\phi,\mu}$ defined by (6), and any two fuzzy measures μ_1 and μ_2 with $\mu_1 < \mu_2$, it has $m_{P,\phi,\mu_1} \leq m_{P,\phi,\mu_2}$.

Remark: Note that for the fuzzy measures $m_{P,\phi,\mu}$ defined by (6), $m_{P,\phi,\mu_*} = m_{P,\phi,\mathrm{Bel}}$ and $m_{P,\phi,\mu^*} = m_{P,\phi,\mathrm{Pl}}$.

The second method will use the constructing paradigm and will be different in the construction choice as compared with what contained in (4) and (5) for constructing the belief and plausibility measure.

Method 2: Generalized the family of fuzzy measures for the belief and plausibility measure.

As discussed in Section II, the source measure space is still $(Q, 2^Q, P)$, and the construction function is still $\phi : Q \to 2^N\backslash\{\varnothing\}$, but the object/constructed measure space becomes $(N, 2^N, m_{P,\phi,\mu})$. The construction choice is based on a known fuzzy measure μ, representing the pessimism–optimism preference of the decision maker and defined such that for any $A \in 2^N\backslash\{\varnothing\}$

$$\begin{aligned} m_{P,\phi,\mu}(A) = {} & P(\{\omega \in Q : \phi(\omega) \subseteq A\}) + \mu(A) \\ & \cdot P(\{\omega \in Q : \phi(\omega) \not\subseteq A,\ \phi(\omega) \cap A \neq \varnothing\}) \end{aligned} \tag{7}$$

and define $m_{P,\phi,\mu}(\varnothing) = 0$.

Remark: About (7), we note that $P(\{\omega \in Q : \phi(\omega) \not\subseteq N, \phi(\omega) \cap N \neq \varnothing\}) = P(\varnothing) = 0$.

Proposition 3: The set function $m_{P,\phi,\mu} : 2^N \to [0,1]$, as defined by (7), is a fuzzy measure.

Proof: 1) We have already defined that $m_{P,\phi,\mu}(\varnothing) = 0$ and

$$\begin{aligned} m_{P,\phi,\mu}(N) &= P(\{\omega \in Q : \phi(\omega) \subseteq N\}) + \mu(N) \\ &\quad \cdot P(\{\omega \in Q : \phi(\omega) \not\subseteq N, \phi(\omega) \cap N \neq \varnothing\}) \\ &= 1 + 1 \cdot P(\varnothing) = 1 + 1 \cdot 0 = 1. \end{aligned}$$

2) For any two subsets $A, B \in 2^N\backslash\{\varnothing\}$ with $A \subset B$, we make the following deduction.

On one hand, first, note that for any $S \in 2^N\backslash\{\varnothing\}$, if $S \not\subseteq B$, then $S \not\subseteq A$. Then, observe that

$$\begin{aligned} & \{\omega \in Q : \phi(\omega) \not\subseteq A, \phi(\omega) \cap A \neq \varnothing\} \\ & \quad = \{\omega \in Q : \phi(\omega) \not\subseteq A, \phi(\omega) \cap A \neq \varnothing, \phi(\omega) \subseteq B\} \\ & \qquad \cup \{\omega \in Q : \phi(\omega) \not\subseteq B, \phi(\omega) \cap A \neq \varnothing\} \end{aligned}$$

and

$$\begin{aligned} & \{\omega \in Q : \phi(\omega) \not\subseteq A, \phi(\omega) \cap A \neq \varnothing,\ \phi(\omega) \subseteq B\} \\ & \quad \cap \{\omega \in Q : \phi(\omega) \not\subseteq B,\ \phi(\omega) \cap A \neq \varnothing\} = \varnothing. \end{aligned}$$

Second, observe that

$$\begin{aligned} \{\omega \in Q : \phi(\omega) \subseteq B\} &= \{\omega \in Q : \phi(\omega) \subseteq A\} \\ &\quad \cup \{\omega \in Q : \phi(\omega) \not\subseteq A, \phi(\omega) \subseteq B\} \end{aligned}$$

and that

$$\{\omega \in Q : \phi(\omega) \subseteq A\} \cap \{\omega \in Q : \phi(\omega) \not\subseteq A, \phi(\omega) \subseteq B\} = \varnothing.$$

One the other hand, observe that

$$\begin{aligned} & \{\omega \in Q : \phi(\omega) \not\subseteq A, \phi(\omega) \cap A \neq \varnothing, \phi(\omega) \subseteq B\} \\ & \quad \subseteq \{\omega \in Q : \phi(\omega) \not\subseteq A, \phi(\omega) \subseteq B\} \end{aligned}$$

and that

$$\begin{aligned} & \{\omega \in Q : \phi(\omega) \not\subseteq B, \phi(\omega) \cap A \neq \varnothing\} \\ & \quad \subseteq \{\omega \in Q : \phi(\omega) \not\subseteq B, \phi(\omega) \cap B \neq \varnothing\}. \end{aligned}$$

Therefore

$$\begin{aligned} m_{P,\phi,\mu}(A) &= P(\{\omega \in Q : \phi(\omega) \subseteq A\}) + \mu(A) \cdot P(\{\omega \in Q : \phi(\omega) \\ &\qquad \not\subseteq A, \phi(\omega) \cap A \neq \varnothing\}) \\ &= P(\{\omega \in Q : \phi(\omega) \subseteq A\}) + \mu(A) \cdot P(\{\omega \in Q : \phi(\omega) \\ &\qquad \not\subseteq A, \phi(\omega) \cap A \neq \varnothing, \phi(\omega) \subseteq B\}) \\ &\qquad + \mu(A) \cdot P(\{\omega \in Q : \phi(\omega) \not\subseteq B, \phi(\omega) \cap A \neq \varnothing\}) \\ &\leq P(\{\omega \in Q : \phi(\omega) \subseteq A\}) + 1 \cdot P(\{\omega \in Q : \phi(\omega) \\ &\qquad \not\subseteq A, \phi(\omega) \subseteq B\}) + \mu(B) \cdot P(\{\omega \in Q : \phi(\omega) \\ &\qquad \not\subseteq B, \phi(\omega) \cap B \neq \varnothing\}) \\ &= P(\{\omega \in Q : \phi(\omega) \subseteq B\}) \\ &\qquad + \mu(B) \cdot P(\{\omega \in Q : \phi(\omega) \not\subseteq B, \phi(\omega) \cap B \neq \varnothing\}) \\ &= m_{P,\phi,\mu}(B). \end{aligned}$$

□

The following monotonicity apparently holds.

Proposition 4: For the fuzzy measures $m_{P,\phi,\mu}$ defined by (7), and any two fuzzy measures μ_1 and μ_2 with $\mu_1 < \mu_2$, it has $m_{P,\phi,\mu_1} \leq m_{P,\phi,\mu_2}$.

Example 1: With $N = \{1,2,3,4\}$, suppose the source measure space $(Q, 2^Q, P)$ and the construction function $\phi : Q \to 2^N\backslash\{\varnothing\}$ are given satisfying

$$\begin{aligned} & P(\{\omega \in Q : \phi(\omega) = \{1\}\}) = 0.1, \\ & P(\{\omega \in Q : \phi(\omega) = \{2\}\}) = 0.1, \\ & P(\{\omega \in Q : \phi(\omega) = \{1,2\}\}) = 0.1, \\ & P(\{\omega \in Q : \phi(\omega) = \{1,4\}\}) = 0.1, \end{aligned}$$

Authorized licensed use limited to: Xi'an Jiaotong-Liverpool University. Downloaded on July 23,2025 at 16:35:24 UTC from IEEE Xplore. Restrictions apply.

$$P(\{\omega \in Q : \phi(\omega) = \{3,4\}\}) = 0.2,$$
$$P(\{\omega \in Q : \phi(\omega) = \{1,2,3\}\}) = 0.2,$$
$$P(\{\omega \in Q : \phi(\omega) = N\}) = 0.2$$

and a decision maker already had a preference in mind, which is expressed as an additive fuzzy measure (also known as probability measure) $\mu : 2^N \to [0,1]$ such that $\mu(\{1\}) = 0.1$, $\mu(\{2\}) = 0.2$, $\mu(\{3\}) = 0.3$, and $\mu(\{4\}) = 0.4$.

Then, by formula (7), we can generate a fuzzy measure $m_{P,\phi,\mu}$ such that

$$m_{P,\phi,\mu}(\{1\}) = (0.1) + (0.1)(0.1+0.1+0.2+0.2) = 0.16,$$
$$m_{P,\phi,\mu}(\{2\}) = (0.1) + (0.2)(0.1+0.1+0.2+0.2) = 0.22,$$
$$m_{P,\phi,\mu}(\{3\}) = 0.18,$$
$$m_{P,\phi,\mu}(\{4\}) = 0.2,$$
$$m_{P,\phi,\mu}(\{1,2\}) = (0.1+0.1+0.1) + (0.3)(0.1+0.2+0.2) = 0.45,$$
$$m_{P,\phi,\mu}(\{1,3\}) = 0.42,$$
$$m_{P,\phi,\mu}(\{1,4\}) = 0.6,$$
$$m_{P,\phi,\mu}(\{2,3\}) = 0.45,$$
$$m_{P,\phi,\mu}(\{2,4\}) = 0.58,$$
$$m_{P,\phi,\mu}(\{3,4\}) = 0.55,$$
$$m_{P,\phi,\mu}(\{1,2,3\}) = 0.8,$$
$$m_{P,\phi,\mu}(\{1,2,4\}) = 0.82,$$
$$m_{P,\phi,\mu}(\{1,3,4\}) = 0.8,$$
$$m_{P,\phi,\mu}(\{2,3,4\}) = 0.84.$$

□

Proposition 5: In light of Method 2 and formula (7), for the source measure space $(Q, 2^Q, P)$ and the construction function $\phi : Q \to 2^N \backslash \{\varnothing\}$, we can have the following.

1) If $P(\{\omega \in Q : \phi(\omega) = N\}) = 1$, then for any already known fuzzy measure μ, we have $m_{P,\phi,\mu} = \mu$.
2) If $\mu = \mu_*$, then $m_{P,\phi,\mu} = m_{P,\phi,\mathrm{Bel}}$.
3) If $\mu = \mu^*$, then $m_{P,\phi,\mu} = m_{P,\phi,\mathrm{Pl}}$.

Proof:

1) $m_{P,\phi,\mu}(\varnothing) = \mu(\varnothing) = 0$, $m_{P,\phi,\mu}(N) = \mu(N) = 1$, and for any $A \in 2^N \backslash \{\varnothing, N\}$, $m_{P,\phi,\mu}(A) = 0 + \mu(A) \cdot 1 = \mu(A)$.

2) $m_{P,\phi,\mu}(\varnothing) = m_{P,\phi,\mathrm{Bel}}(\varnothing) = 0$, $m_{P,\phi,\mu}(N) = m_{P,\phi,\mathrm{Bel}}(N) = 1$, and for any $A \in 2^N \backslash \{\varnothing, N\}$, $m_{P,\phi,\mu}(A) = P(\{\omega \in Q : \phi(\omega) \subseteq A\}) = m_{P,\phi,\mathrm{Bel}}(A)$.

3) $m_{P,\phi,\mu}(\varnothing) = m_{P,\phi,\mathrm{Pl}}(\varnothing) = 0$, $m_{P,\phi,\mu}(N) = m_{P,\phi,Pl}(N) = 1$, and for any $A \in 2^N \backslash \{\varnothing, N\}$, $m_{P,\phi,\mu}(A) = P(\{\omega \in Q : \phi(\omega) \cap A \neq \varnothing\}) = m_{P,\phi,\mathrm{Pl}}(A)$. □

IV. To Determine Construction Functions From Known Probability Measure

In Section III, we have shown that in the risk evaluation and decision making, the pessimism–optimism preferences of decision makers may affect or decide the process of construction choice in constructing paradigm. Actually, the construction function made, for example, by experts can also be affected or decided by the pessimism–optimism preferences of those experts in some certain decision situations.

Let us assume the provided measure space in the risk evaluation is a probability space $(N, 2^N, \tilde{P})$, but the decision maker does not believe that the probability measure $\tilde{P}$ can fully reflect the real situation of the states of nature. Therefore, he/she may use the constructing paradigm to construct a new measure space to replace it. Suppose the invited q experts have the varying extents of bipolar pessimism–optimism preferences over the provided probability measure $\tilde{P}$, and the extents are expressed simply by q different real numbers in unit interval [0,1], $\lambda_j \in [0,1]$ $(j \in Q = \{1, \ldots, q\})$, respectively. We suppose a larger λ_j expressed by some experts represents a larger extent of the pessimism of him/her, and a smaller of λ_j indicates a larger optimism of him/her.

In what follows, we will discuss three different methods to automatically obtain the construction functions, $\phi : Q \to 2^N/\{\varnothing\}$, extracted from the given probability space $(N, 2^N, \tilde{P})$ and the pessimism–optimism preferences $\{\lambda_j\}_{j=1}^q$ communicated by experts.

Note that the tied values in $\tilde{P}(\{i\})$ $(i \in N)$ may significantly complicate the expressions of the extraction processes, making it very hard for the practitioners to well program. Thus, for the following three methods, we elaborately devise some strict, accurate, and workable formulating techniques without which the possible tied values in $\tilde{P}(\{i\})$ might spoil some attempts to correctly formulate the problem.

Construction function extraction—Method 1:

Step 1: For the given probability space $(N, 2^N, \tilde{P})$, define $\max_{i \in N}(\tilde{P}(\{i\})) = M$. Clearly, $0 < M \leq 1$.

Step 2: Define function $f_{\tilde{P},1} : [0,1] \to 2^N \backslash \{\varnothing\}$ such that for $t \in (0,1]$

$$f_{\tilde{P},1}(t) = \{s \in N : \tilde{P}(\{s\}) \geq t \cdot M\} \tag{8}$$

and

$$f_{\tilde{P},1}(0) = \{s \in N : \tilde{P}(\{s\}) > 0\}. \tag{9}$$

Step 3: Define a construction function $\phi_1 : Q \to 2^N \backslash \{\varnothing\}$ such that

$$\phi_1(j) = f_{\tilde{P},1}(\lambda_j). \tag{10}$$

We have the following two remarks.

1) The reason why we specially define $\phi_1(j) = \{s \in N : \tilde{P}(\{s\}) > 0\}$ when $\lambda_j = 0$ is because we take the cognitive assumption that if $\tilde{P}(\{i\}) = 0$, then any expert involved will also think the state of nature *i* is impossible to occur. This assumption will be consistent throughout all of the three methods proposed in this section. However, the readers can slightly adapt the proposed methods according to different situations and cognitive assumptions.
2) When $\lambda_j = 1$, representing an extreme pessimism of expert *j*, then $\phi_1(j)$ only indicates the states of nature that are associated with the biggest probabilities.

Construction function extraction—Method 2:

Step 1: For the given probability space $(N, 2^N, \tilde{P})$, define $\max_{i \in N}(\tilde{P}(\{i\})) = M$. Clearly, $0 < M \leq 1$.

Step 2: Define a continuous, monotonic increasing and surjective function $g_{\tilde{P}} : [0, M] \to [0,1]$ such that

$$g_{\tilde{P}}(z) = 1 - \sum_{k \in \{s \in N : \tilde{P}(\{s\}) \geq z\}} \left(\tilde{P}(\{k\}) - z\right). \tag{11}$$

Step 3: Define function $f_{\tilde{P},2} : [0,1] \to 2^N \backslash \{\varnothing\}$ such that for $t \in (0,1]$

$$f_{\tilde{P},2}(t) = \{s \in N : \tilde{P}(\{s\}) \geq {g_{\tilde{P}}}^{-1}(t)\} \tag{12}$$

and

$$f_{\tilde{P},2}(0) = \{s \in N : \tilde{P}(\{s\}) > 0\}. \tag{13}$$

Step 4: Define the construction function $\phi_2 : Q \to 2^N \backslash \{\varnothing\}$ such that

$$\phi_2(j) = f_{\tilde{P},2}(\lambda_j). \tag{14}$$

Authorized licensed use limited to: Xi'an Jiaotong-Liverpool University. Downloaded on July 23,2025 at 16:35:24 UTC from IEEE Xplore. Restrictions apply.

We have the following two remarks.

1) $g_{\tilde{P}}(0) = 0$ and $g_{\tilde{P}}(M) = 1$.
2) Since *n* is finite, then for any $t \in (0, 1]$, finding $g_{\tilde{P}}{}^{-1}(t)$ can be easily achieved with finite times of simple enumerations, which will not be discussed further in this study.

Proposition 6: In the construction function extraction—Method 1 and Method 2, $f_{\tilde{P},1}(t) \subseteq f_{\tilde{P},2}(t)$ for any $t \in [0, 1]$.

Proof:

1) $f_{\tilde{P},1}(0) = f_{\tilde{P},2}(0)$.

2) From (8), (11), and (12), for any $t \in (0, 1]$, note that for each $k \in \{s \in N : \tilde{P}(\{s\}) \geq tM\}$, $t = \frac{tM}{M} \leq \frac{tM}{\tilde{P}(\{k\})}$, and for each $k \in \{s \in N : 0 < \tilde{P}(\{s\}) < tM\}$, $t \leq \frac{\tilde{P}(\{k\})}{\tilde{P}(\{k\})} = 1$. Therefore

$$\begin{aligned} & g_{\tilde{P}}\left(g_{\tilde{P}}{}^{-1}(t)\right) \\ &= t \leq \frac{\sum\limits_{k\in\{s\in N:\tilde{P}(\{s\})\geq tM\}} tM + \sum\limits_{k\in\{s\in N:0<\tilde{P}(\{s\})<tM\}} \tilde{P}(\{k\})}{\sum\limits_{k\in\{s\in N:\tilde{P}(\{s\})\geq tM\}} \tilde{P}(\{k\}) + \sum\limits_{k\in\{s\in N:0<\tilde{P}(\{s\})<tM\}} \tilde{P}(\{k\})} \\ &= \frac{1 - \sum\limits_{k\in\{s\in N:\tilde{P}(\{s\})\geq tM\}} \left(\tilde{P}(\{k\}) - tM\right)}{1} = g_{\tilde{P}}(t \cdot M). \end{aligned}$$

Since $g_{\tilde{P}}$ is increasing, then $g_{\tilde{P}}{}^{-1}(t) \leq t \cdot M$, which indicates $f_{\tilde{P},1}(t) \subseteq f_{\tilde{P},2}(t)$. □

Construction function extraction—Method 3:

Step 1: Obtain a function $p : N \to [0, 1]$ from probability measure $\tilde{P}$ such that $p(i) = \tilde{P}(\{i\})$. Clearly, $\sum_{i=1}^{n} p(i) = 1$.

Step 2: Define a number set $\{a_k\}_{k=1}^{H}$ such that $a_k > a_{k+1}$ for all $k \in \{1, \ldots, H-1\}$ and $\{a_k\}_{k=1}^{H} = \mathrm{Ran}(p)$, where $\mathrm{Ran}(p)$ is the range of $p : N \to [0, 1]$. Clearly, $1 \leq H \leq n$ and $a_1 > 0$.

Step 3: Define function $r : [0, 1] \to \{1, \ldots, H\}$ such that for any $t \in [0, 1]$, $r(t)$ simultaneously satisfies the following two conditions:

1) $\sum\limits_{k\in\{s\in N:p_s\geq a_{r(t)}\}} p_k \geq t$;
2)
$$\sum_{k\in\{s\in N:p_s\geq a_{r(t)-1}\}} p_k < t \text{ (we define } a_0 = -1). \tag{15}$$

Step 4: Define function $f_{\tilde{P},3} : [0, 1] \to 2^N \backslash \{\varnothing\}$ such that for $t \in [0, 1]$

$$f_{\tilde{P},3}(t) = \{s \in N : p_s \geq a_{r(1-t)}\}. \tag{16}$$

Step 5: Define the construction function $\phi_3 : Q \to 2^N \backslash \{\varnothing\}$ such that

$$\phi_3(j) = f_{\tilde{P},3}(\lambda_j). \tag{17}$$

We have the following two remarks.

1) We observe that if $\lambda_j = 0$, then $\phi_3(j) = \{s \in N : \tilde{P}(\{s\}) > 0\}$.
2) If $\tilde{P}(\{i\}) = p(i) = 1/n$ for any $i \in N$, then for any $\lambda_j \in [0, 1]$, $\phi_3(j) = N$.

Remark: For the functions $f_{\tilde{P},k}$ $(k = 1, 2, 3)$ defined, respectively, in the above proposed three methods

1) The function $f_{\tilde{P},k}$ $(k = 1, 2, 3)$ is monotonic; that is, if $x < y$, then $f_{\tilde{P},k}(y) \subseteq f_{\tilde{P},k}(x)$, indicating that the bipolar pessimism–optimism preferences of experts are well and consistently modeled.
2) With any two states of nature *i* and *j*, if $\tilde{P}(\{i\}) < \tilde{P}(\{j\})$, then for any $t \in [0, 1]$, $i \in f_{\tilde{P},k}(t) \Rightarrow j \in f_{\tilde{P},k}(t)$ $(k = 1, 2, 3)$. This indicates that if a state of nature with smaller originally provided probability is considered to be going to occur, then those states of natures with larger probabilities will also occur, which conforms to human usual intuition.

Example 2: Suppose the given probability space $(N, 2^N, \tilde{P})$ $(N = \{1, 2, 3, 4, 5\})$ is with probability measure $\tilde{P} : 2^N \to [0, 1]$, which is not fully believed by the decision maker and such that $\tilde{P}(\{1\}) = \tilde{P}(\{2\}) = 0.25$, $\tilde{P}(\{3\}) = 0.4$, $\tilde{P}(\{4\}) = 0$, and $\tilde{P}(\{5\}) = 0.1$.

Assume $Q = \{1, 2, 3\}$ and $\lambda_1 = 0.2$, $\lambda_2 = 0.5$, and $\lambda_3 = 0.7$ representing three invited experts and indicating their pessimism extents, respectively.

With the construction function extraction—Method 1, we have $M = 0.4$ and obtain construction function $\phi_1 : Q \to 2^N \backslash \{\varnothing\}$ such that

$$\phi_1(1) = f_{\tilde{P},1}(0.2) = \{s \in N : \tilde{P}(\{s\}) \geq 0.08\} = \{1, 2, 3, 5\},$$

$$\phi_1(2) = f_{\tilde{P},1}(0.5) = \{s \in N : \tilde{P}(\{s\}) \geq 0.2\} = \{1, 2, 3\},$$

$$\phi_1(3) = f_{\tilde{P},1}(0.7) = \{s \in N : \tilde{P}(\{s\}) \geq 0.28\} = \{3\}.$$

With the construction function extraction—Method 2, we have $M = 0.4$ and obtain construction function $\phi_1 : Q \to 2^N \backslash \{\varnothing\}$ such that

$$\phi_2(1) = f_{\tilde{P},2}(0.2) = \{s \in N : \tilde{P}(\{s\}) \geq 0.05\} = \{1, 2, 3, 5\},$$

$$\phi_2(2) = f_{\tilde{P},2}(0.5) = \{s \in N : \tilde{P}(\{s\}) \geq 2/15\} = \{1, 2, 3\},$$

$$\phi_2(3) = f_{\tilde{P},2}(0.7) = \{s \in N : \tilde{P}(\{s\}) \geq 0.2\} = \{1, 2, 3\}.$$

With the construction function extraction—Method 2, we have $\{a_k\}_{k=1}^{H} = \{a_k\}_{k=1}^{4} = \{0.4, 0.25, 0.1, 0\}$ and

$$\begin{aligned} \phi_3(1) &= f_{\tilde{P},3}(0.2) = \{s \in N : p_s \geq a_{r(0.8)}\} \\ &= \{s \in N : p_s \geq a_2\} = \{s \in N : p_s \geq 0.25\} = \{1, 2, 3\}, \\ \phi_3(2) &= f_{\tilde{P},3}(0.5) = \{s \in N : p_s \geq a_{r(0.5)}\} \\ &= \{s \in N : p_s \geq a_2\} = \{s \in N : p_s \geq 0.25\} = \{1, 2, 3\}, \\ \phi_3(3) &= f_{\tilde{P},3}(0.7) = \{s \in N : p_s \geq a_{r(0.3)}\} \\ &= \{s \in N : p_s \geq a_1\} = \{s \in N : p_s \geq 0.4\} = \{1\}. \end{aligned}$$ □

V. Conclusion

With some theoretical derivation and practical evaluation problems, in this article, we proposed a constructing paradigm, which can help decision makers to construct fuzzy measures with a strict and convenient formulation. It elaborately analyzed and dissected a type of fuzzy measures constructing problems into four factors, including source measure space, construction function, construction choice, and object/constructed measure space.

As two special instances of the proposed paradigm, we illustrated the way to construct the well-known belief measure and plausibility measure. Furthermore, we proposed two different construction choices to form two families of fuzzy measures with given preferences, which are expressed by known fuzzy measures. Both of the two families of fuzzy measures have the desired monotonicities and can well generalize and seamlessly connect the belief measure and plausibility measure.

For a practical problem in the risk evaluation, we elaborately proposed three different methods to generate construction functions. The possible tied values in the problem forced us to carefully devise and well formulate the generating and extracting processes. The methods were suitable when the probability measures needed to be used in the evaluation cannot fully reflect the real situation of the states of nature. The pessimism–optimism preferences of experts also affected the generated construction functions.

Authorized licensed use limited to: Xi'an Jiaotong-Liverpool University. Downloaded on July 23,2025 at 16:35:24 UTC from IEEE Xplore. Restrictions apply.

The proposed paradigm can have more ramifications and instances, and more methods to determine the construction function and construction choice can be found in different applications, which will be further studied in later works.

Acknowledgment

The authors would like to thank the constructive comments from reviewers and editors.

References

[1] R. R. Yager, "A fuzzy measure approach to systems reliability modeling," *IEEE Trans. Fuzzy Syst.*, vol. 22, no. 5, pp. 1139–1150, Oct. 2014.

[2] D. Paternain, J. Fernandez, H. Bustince, R. Mesiar, and G. Beliakov, "Construction of image reduction operators using averaging aggregation functions," *Fuzzy Set Syst.*, vol. 261, pp. 87–111, 2015.

[3] T. Leon, P. Zuccarello, G. Ayala, E. de Ves, and J. Domingo, "Applying logistic regression to relevance feedback in image retrieval systems," *Pattern Recognit.*, vol. 40, pp. 2621–2632, 2007.

[4] J. E. Hirsch, "An index to quantify an individual's scientific research output," *Proc. Nat. Acad. Sci.*, vol. 102, no. 46, pp. 16569–16572, 2005.

[5] L. Jin, R. Mesiar, and A. Stupnanova, "Sugeno integrals, H-alpha and H-beta indices: How to compare scientists from different academic areas," *IEEE Trans. Fuzzy Syst.*, vol. 28, no. 4, pp. 795–800, Apr. 2020.

[6] S. Bodjanova and M. Kalina, "Approximate evaluations based on aggregation functions," *Fuzzy Sets Syst.*, vol. 220, no. 1, pp. 34–52, 2013.

[7] X. Liu and Q. Da, "A decision tree solution considering decision maker's attitude," *Fuzzy Sets Syst.*, vol. 152, pp. 437–454, 2005.

[8] C.-L. Hwang and K. Yoon, *Multiple Attribute Decision Making*. Berlin, Germany: Springer, 1981.

[9] R. L. Keeney and H. Raiffa, *Decisions With Multiple Objectives: Preferences and Value Tradeoffs*. Hoboken, NJ, USA: Wiley, 1976.

[10] S. M. Chen and W. Pedrycz, *Granular Computing and Decision-Making: Interactive and Iterative Approaches*. Berlin, Germany: Springer, 2015.

[11] R. R. Yager, "On ordered weighted averaging aggregation operators in multicriteria decision making," *IEEE Trans. Syst. Man Cybern.*, vol. 18, no. 1, pp. 183–190, Jan./Feb. 1988.

[12] Y. Even and E. Lehrer, "Decomposition-integral: Unifying Choquet and the concave integrals," *Econ. Theory*, vol. 56, pp. 33–58, 2014.

[13] B. Llamazares, "Choosing OWA operator weights in the field of social choice," *Inf. Sci.*, vol. 177, pp. 4745–4756, 2007.

[14] J. Malczewski, "Ordered weighted averaging with fuzzy quantifiers: GIS-based multicriteria evaluation for land-use suitability analysis," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 8, pp. 270–277, 2006.

[15] T. L. Saaty, "Axiomatic foundation of the analytic hierarchy process," *Manage. Sci.*, vol. 32, no. 7, pp. 841–855, 1986.

[16] J. Li, X. Yao, X. Sun, and D. Wu, "Determining the fuzzy measures in multiple criteria decision aiding from the tolerance perspective," *Eur. J. Oper. Res.*, vol. 264, pp. 428–439, 2018.

[17] X. Liu and S. Han, "Orness and parameterized RIM quantifier aggregation with OWA operators: A summary," *Int. J. Approx. Reasoning*, vol. 48, pp. 77–97, 2008.

[18] L. Jin, R. Mesiar, and G. Qian, "Weighting models to generate weights and capacities in multicriteria group decision making," *IEEE Trans. Fuzzy Syst.*, vol. 26, no. 4, pp. 2225–2236, Aug. 2018.

[19] L. Jin, R. Mesiar, and R. R. Yager, "Melting probability measure with OWA operator to generate fuzzy measure: The crescent method," *IEEE Trans. Fuzzy Syst.*, vol. 27, no. 6, pp. 1309–1316, Jun. 2019.

[20] L. Jin, M. Kalina, R. Mesiar, and S. Borkotokey, "Certainty aggregation and the certainty fuzzy measures," *Int. J. Intell. Syst.*, vol. 33, no. 4, pp. 759–770, 2018.

[21] B. Llamazares, "A behavioral analysis of WOWA and SUOWA operators," *Int. J. Intell. Syst.*, vol. 31, no. 8, pp. 827–851, 2016.

[22] J.-L. Marichal, "Tolerant or intolerant character of interacting criteria in aggregation by the Choquet integral," *Eur. J. Oper. Res.*, vol. 155, no. 3, pp. 771–791, 2004.

[23] Y. Ouyang, "Improved minimax disparity model for obtaining OWA operator weights: Issue of multiple solutions," *Inf. Sci.*, vol. 320, no. 1, pp. 101–106, 2015.

[24] M. Grabisch, J.-L. Marichal, R. Mesiar, and E. Pap, *Aggregation Functions*. Cambridge, U.K.: Cambridge Univ. Press, 2009.

[25] M. Sugeno, "Theory of fuzzy integrals and its applications," Ph.D. dissertation, Tokyo Inst. Technol., Tokyo, Japan, 1974.

[26] G. Choquet, "Theory of capacities," *Ann. Inst. Fourier*, vol. 5, pp. 131–295, 1954.

[27] N. Shilkret, "Maxitive measure and integration," *Indagationes Math.*, vol. 74, pp. 109–116, 1971.

[28] Q. Yang, "The pan-integral on fuzzy measure space," *Fuzzy Math.*, vol. 3, pp. 107–114, 1985.

[29] R. Mesiar and A. Stupňanová, "Decomposition integrals," *Int. J. Approx. Reasoning*, vol. 54, pp. 1252–1259, 2013.

[30] L. Jin, R. Mesiar, and R. R. Yager, "Derived fuzzy measures and derived Choquet integrals with some properties," *IEEE Trans. Fuzzy Syst.*, early access, doi: 10.1109/TFUZZ.2020.2969869.

[31] E. P. Klement, R. Mesiar, and E. Pap, "A universal integral as common frame for Choquet and Sugeno integral," *IEEE Trans. Fuzzy Syst.*, vol. 18, no. 1, pp. 178–187, Feb. 2010.

[32] L. Jin, M. Kalina, R. Mesiar, and S. Borkotokey, "Discrete Choquet integrals for Riemann integrable inputs with some applications," *IEEE Trans. Fuzzy Syst.*, vol. 26, no. 5, pp. 3164–3169, Oct. 2018.

[33] X. Pu, R. Mesiar, R. R. Yager, and L. Jin, "Interval Sugeno integral with preference," *IEEE Trans. Fuzzy Syst.*, vol. 28, no. 3, pp. 597–601, Mar. 2020.

[34] E. P. Klement, R. Mesiar, and E. Pap, *Triangular Norms*. Dordrecht, The Netherlands: Springer, 2000.

[35] Y. Ouyang, "On fuzzy implications determined by aggregation operators," *Inf. Sci.*, vol. 193, no. 15, pp. 153–162, 2012.

[36] R. R. Yager, J. Kacprzyk, and G. Beliakov, *Recent Developments in the Ordered Weighted Averaging Operators: Theory and Practice*. Berlin, Germany: Springer, 2011.

[37] T. Murofushi and M. Sugeno, "An interpretation of fuzzy measures and the Choquet integral as an integral with respect to a fuzzy measure," *Fuzzy Sets Syst.*, vol. 29, pp. 201–227, 1989.

[38] R. Mesiar, S. Borkotokey, L. Jin, and M. Kalina, "Aggregation functions and capacities," *Fuzzy Sets Syst.*, vol. 346, no. 1, pp. 138–146, 2018.

[39] V. Torra and Y. Narukawa, "The interpretation of fuzzy integrals and their application to fuzzy systems," *Int. J. Approx. Reasoning*, vol. 41, pp. 43–58, 2006.

[40] N. M. Ralevic and M. Paunovic, "C-credibility measure," *Filomat*, vol. 33, no. 9, pp. 2571–2582, 2019.

[41] C. Zhu, L. Jin, R. Mesiar, and R. R. Yager, "Using preference leveled evaluation functions to construct fuzzy measures in decision making and evaluation," *Int. J. Gen. Syst.*, vol. 49, no. 2, pp. 161–173, 2020.

[42] A. Papoulis, *Probability, Random Variables, and Stochastic Processes*, 3rd ed. New York, NY, USA: McGraw-Hill, 1991.

[43] A. P. Dempster, "Upper and lower probabilities induced by a multivalued mapping," *Ann. Math. Statist.*, vol. 38, no. 2, pp. 325–339, 1967.

[44] G. Shafer, *A Mathematical Theory of Evidence*. Princeton, NJ, USA: Princeton Univ. Press, 1976.

Authorized licensed use limited to: Xi'an Jiaotong-Liverpool University. Downloaded on July 23,2025 at 16:35:24 UTC from IEEE Xplore. Restrictions apply.